Week 8 — Robustness Output (Edge Cases)

Goal

Handle edge cases gracefully and log warnings in the trail.

Required Outputs

☐ Edge case list (minimum 5)

☐ At least 3 edge-case tests documented

☐ No crashes; warnings/messages shown in UI

☐ Trail logs warnings clearly

Evidence to Attach

☐ Screenshots: 3 edge-case runs (UI + trail)

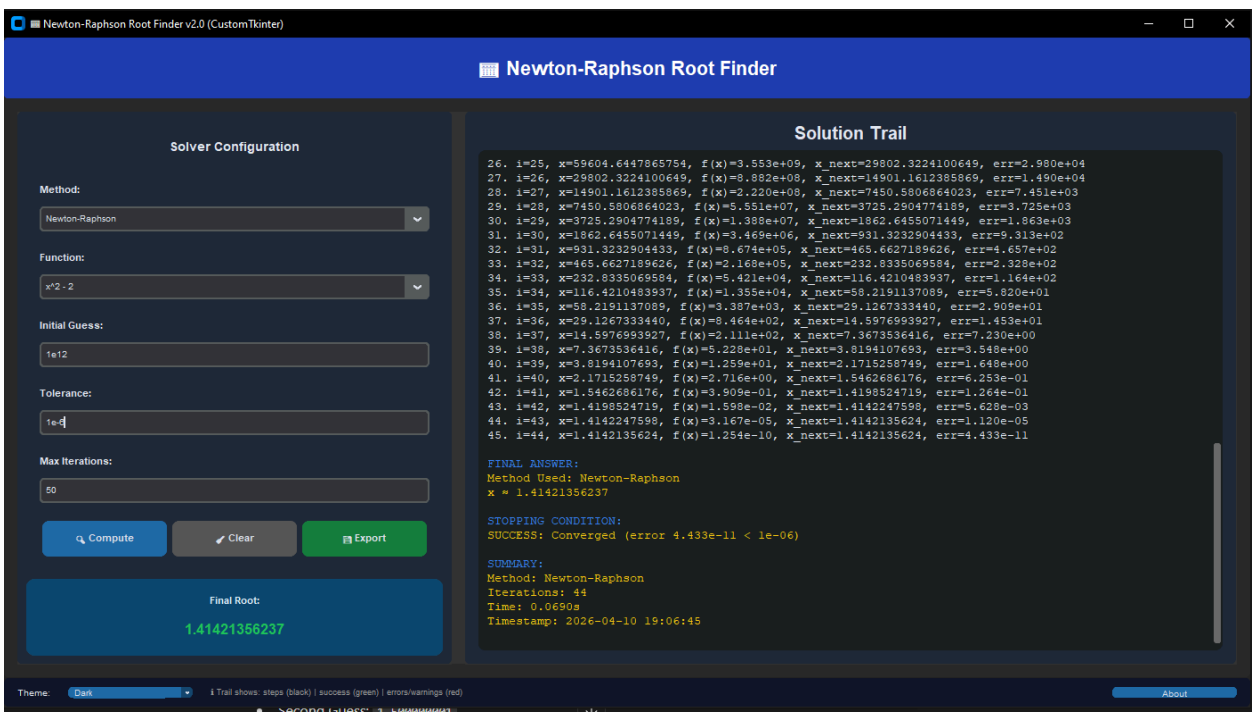
Reflection (Write short answers)

- Which edge case produced the most bugs and why?

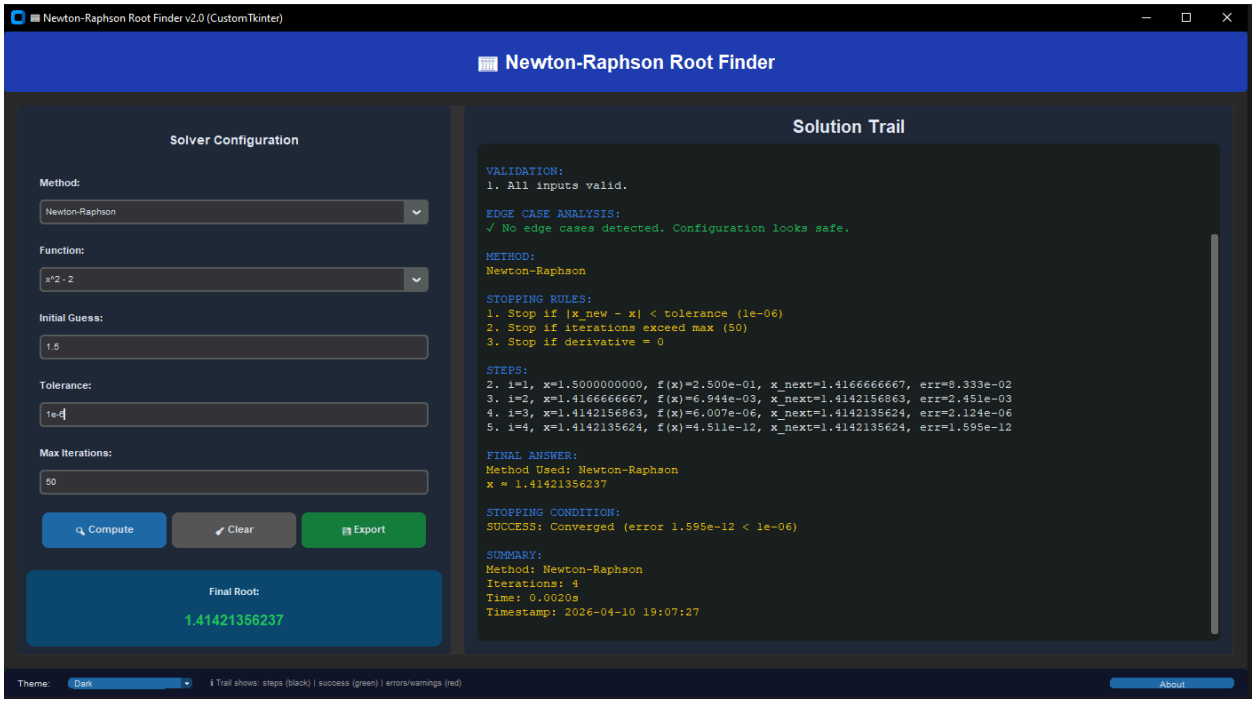
Edge Cases Identified:

1. Extremely large initial guess
2. Extremely small tolerance
3. Very low max iterations
4. Equal guesses in Secant method
5. Derivative near zero

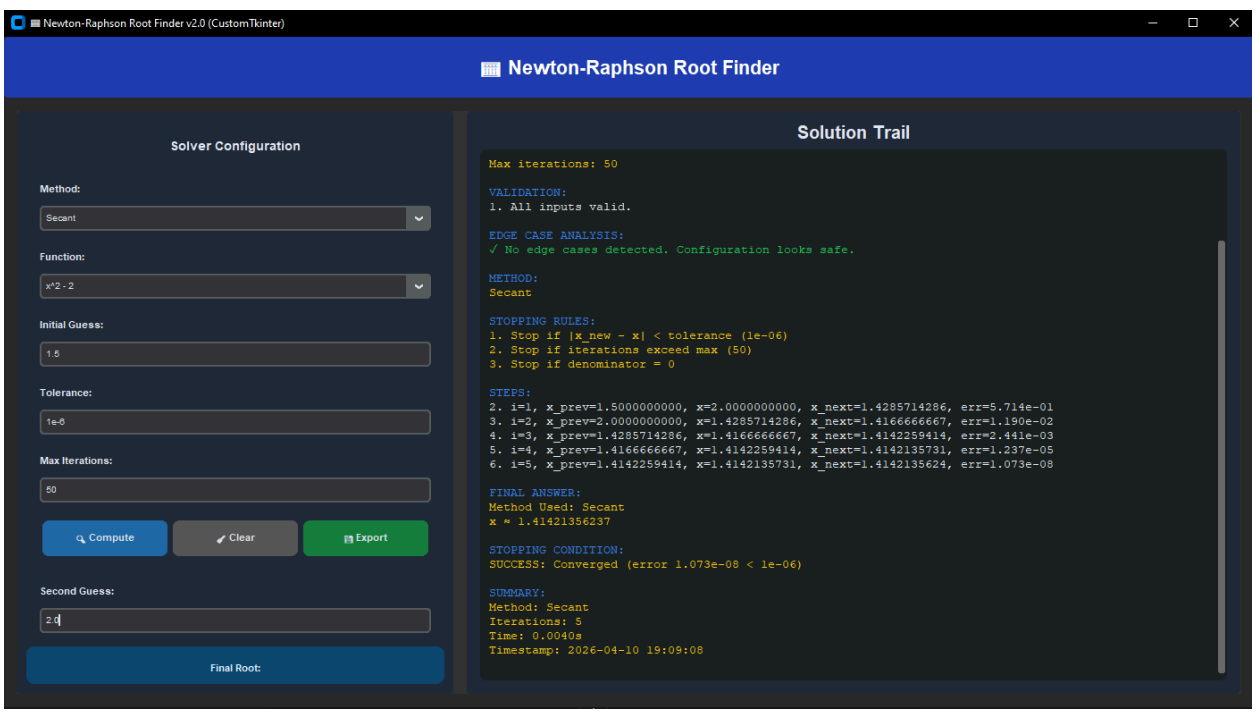
TEST #1 Large Initial Guess



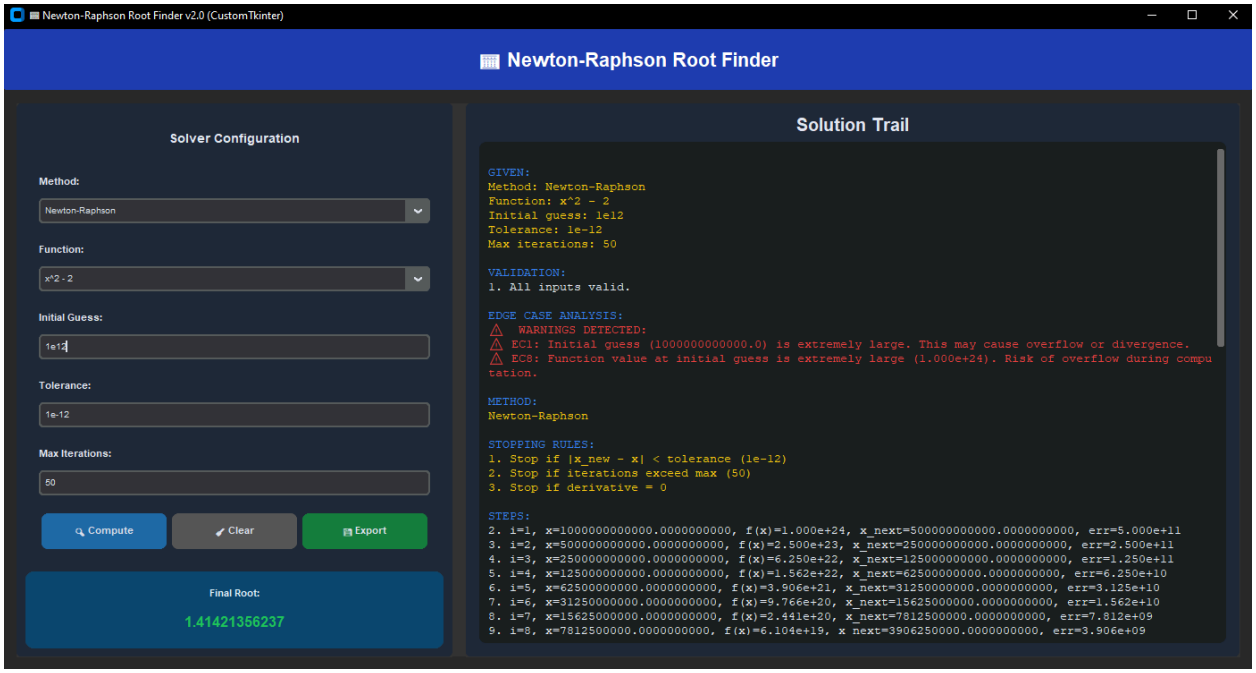
TEST#2 Small Tolerance

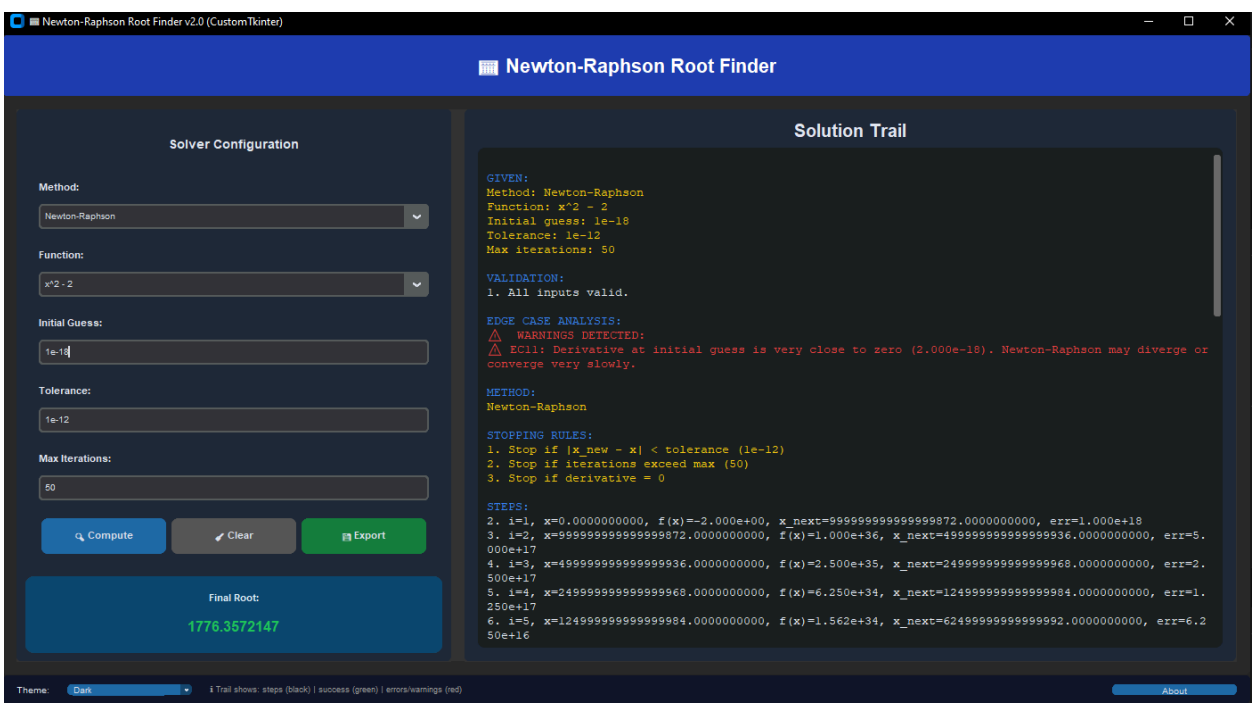


TEST#3 Secant Edge Case



Trail logs warnings clearly





Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Secant
Function:
x^2 - 2
Initial Guess:
1.5
Tolerance:
1e-12
Max Iterations:
50
Compute
Clear
Export
Second Guess:
1.5
Final Root:
Solution Trail
Tolerance: 1e-12
Max iterations: 50
VALIDATION:
1. All inputs valid.
EDGE CASE ANALYSIS:
WARNINGS DETECTED:
EC6: Secant method: initial guesses are too close (0.00e+00). Denominator may become zero. Use more different guesses.
METHOD:
Secant
STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-12)
2. Stop if iterations exceed max (50)
3. Stop if denominator = 0
STEPS:
2. STOP: Zero denominator at iteration 1
WARNING: f(x) - f(x_prev) = 0. Secant method cannot continue.
FINAL ANSWER:
Method Used: Secant
x ≈ 1.5
STOPPING CONDITION:
STOP: Zero denominator at iteration 1
SUMMARY:
Method: Secant
Iterations: 1
Time: 0.0010s
Timestamp: 2026-04-10 19:40:07

GITHUB REPO

Reflection

Which edge case produced the most bugs and why?

- The Secant method edge case with nearly equal initial guesses ($x_0 \approx x_1$) produced the most bugs. This is because the method relies on the difference (f(x) - f(x_{prev})) as the denominator, and when the two guesses are very close, this value becomes extremely small or zero. As a result, it caused division-by-zero errors or unstable large jumps in the computed next value.
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- This edge case was particularly difficult to handle because it does not always immediately produce an obvious error—sometimes it leads to incorrect results instead of a crash. To fix this, additional validation and stopping conditions were implemented to detect near-zero denominators and safely terminate the computation while logging a clear warning in the solution trail.

Weekly Sign-off

Group Leader Name/Signature: CHARLES RUSSEL O. SAN JUAN Date: 4/7/2026

Member Name/Signature: CHRISTINA GOMBA Date: 4/7/2026

Member Name/Signature: JOHN MICHAEL TULALIAN Date: 3/7/2026